

Approximation Hardness of Assortment Optimization under the Consideration Set Model

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Abstract

We study assortment optimization under the consideration set model, in which each customer chooses uniformly among the offered products in her consideration set. For every fixed $\varepsilon \in (0, 1)$, we prove that no deterministic polynomial-time algorithm can approximate the optimal expected revenue within a factor of $O(n^{1-\varepsilon})$ unless $P = NP$, where n is the number of products. This improves the $O(n^{1/2-\varepsilon})$ hardness bound of Akchen and Mitrofanov. We use their reduction from Maximum Independent Set and give a new extraction argument based on first-fit coloring. In the constructed instance, any nonempty assortment can be converted into an independent set whose cardinality is at least the assortment's revenue divided by $H_n + 1/n$, where H_n is the n th harmonic number. The hardness holds with n distinct consideration sets and rational parameters of polynomial encoding length. Together with the elementary n -approximation obtained by offering all products, the result determines the polynomial exponent of the approximation factor. These bounds leave open improvements by logarithmic or other subpolynomial factors.

1 Introduction

Assortment optimization asks which products a seller should offer to maximize expected revenue under a given model of customer choice. The choice model matters because offering an additional product can change the demand for products already available. In the consideration set model, a customer is associated with a set of products she is willing to purchase. She chooses uniformly from the intersection of this set with the offered assortment, and makes no purchase if the intersection is empty. Even in this simple model, adding a product can decrease revenue by reducing the probability of selling a more valuable product.

Akchen and Mitrofanov [1] study this model and show that its assortment optimization problem is NP-hard to approximate within a factor of $O(n^{1/2-\varepsilon})$ for every fixed $\varepsilon \in (0, 1/2)$. Offering all n products, on the other hand, gives an n -approximation. They leave the gap between these bounds as an open question. Their hardness construction is based on an earlier reduction of Aouad, Farias, Levi, and Segev [2] for assortment optimization under ranking preferences. In the ranking model, a customer purchases her most preferred available product. The extraction used for that model relies on this preference order and does not give the same guarantee when the customer instead chooses uniformly.

We improve the hardness bound to $O(n^{1-\varepsilon})$ for every fixed $\varepsilon \in (0, 1)$. The reduction itself is unchanged. The improvement comes from a different way to extract an independent set from an assortment: we color the corresponding vertices greedily and return a largest color class. Summing reciprocal color indices gives a harmonic bound on the loss in this conversion. Combined with the deterministic hardness theorem of Zuckerman [3], this establishes the claimed bound under $P \neq NP$.

1.1 Model and approximation convention

Let $[n] = \{1, \dots, n\}$ denote the set of products. Product i has revenue $r_i > 0$. There are m customer types. Type $t \in [m]$ occurs with probability $\lambda_t > 0$ and has a consideration set $C_t \subseteq [n]$, with

$$\sum_{t=1}^m \lambda_t = 1.$$

An assortment is a subset $S \subseteq [n]$. Conditional on the customer being of type t , the expected revenue from offering S is

$$R_t(S) = \begin{cases} \frac{\sum_{i \in C_t \cap S} r_i}{|C_t \cap S|}, & C_t \cap S \neq \emptyset, \\ 0, & C_t \cap S = \emptyset. \end{cases} \quad (1)$$

The seller's expected revenue and optimal value are, respectively,

$$R(S) = \sum_{t=1}^m \lambda_t R_t(S), \quad \text{OPT} = \max_{S \subseteq [n]} R(S). \quad (2)$$

There is no cardinality constraint on S .

The input explicitly lists the consideration sets, their probabilities, and the product revenues. All probabilities and revenues are rational numbers encoded in binary. Polynomial time is measured in the total input length. An approximation factor, however, is expressed as a function of the number of products n . For a function $\rho(n) \geq 1$, a $\rho(n)$ -approximation algorithm returns an assortment S satisfying

$$R(S) \geq \frac{\text{OPT}}{\rho(n)} \quad (3)$$

on every input. Thus a smaller approximation factor is better.

1.2 Main result

Theorem 1. *For every fixed $\varepsilon \in (0, 1)$, there is no deterministic polynomial-time $O(n^{1-\varepsilon})$ -approximation algorithm for assortment optimization under the consideration set model unless $P = NP$. This holds even when $m = n$ and the n consideration sets are pairwise distinct.*

The theorem rules out, in particular, an $O(\sqrt{n})$ -approximation. Its dependence on n matches the following elementary guarantee at the level of the polynomial exponent.

Proposition 2. *Offering all products is an n -approximation for the consideration set model.*

Proof. Fix an assortment S and a customer type t . If $C_t \cap S$ is empty, then $R_t(S) = 0$ and $R_t([n]) \geq R_t(S)/n$. Otherwise C_t is nonempty, and

$$R_t([n]) = \frac{\sum_{i \in C_t} r_i}{|C_t|} \geq \frac{\sum_{i \in C_t \cap S} r_i}{|C_t|} \geq \frac{\sum_{i \in C_t \cap S} r_i}{n} \geq \frac{1}{n} \frac{\sum_{i \in C_t \cap S} r_i}{|C_t \cap S|} = \frac{R_t(S)}{n}.$$

The inequalities use, in order, nonnegative revenues, $|C_t| \leq n$, and $|C_t \cap S| \geq 1$. Multiplying by λ_t and summing over all types gives

$$R([n]) = \sum_{t=1}^m \lambda_t R_t([n]) \geq \frac{1}{n} \sum_{t=1}^m \lambda_t R_t(S) = \frac{R(S)}{n}.$$

Taking S to be an optimal assortment proves the claim. $\square$

1.3 Proof outline

The reduction associates a product and a customer type with each vertex of an ordered graph. The consideration set of vertex i contains i and its earlier neighbors. The probability and revenue parameters are chosen so that, when product i is offered, the contribution from type i purchasing that product is

$$\frac{1}{1 + d_S^-(i)},$$

where $d_S^-(i)$ counts the earlier neighbors of i in the assortment S . Purchases by type i of products $j < i$ contribute at most $1/n$ in total over all customer types.

The main step is to relate the sum of these reciprocal quantities to a computable independent set. First-fit coloring in the same vertex order assigns to each vertex a color no larger than one plus its number of earlier neighbors. Grouping vertices by color then yields a harmonic bound. Section 2 proves this graph inequality. Section 3 gives the reduction and the revenue estimates. Section 4 combines them to prove Theorem 1.

2 A harmonic bound for ordered graphs

Let $G = ([n], E)$ be a simple undirected graph, with the vertex order $1, \dots, n$. An independent set is a subset containing no two adjacent vertices. We write $\alpha(G)$ for the maximum cardinality of an independent set in G , and $G[S]$ for the subgraph induced by $S \subseteq [n]$.

For $S \subseteq [n]$ and $i \in S$, define the number of earlier neighbors of i in S by

$$d_S^-(i) = |\{j \in S : j < i \text{ and } \{i, j\} \in E\}|. \quad (4)$$

The quantity of interest is

$$F_G(S) = \sum_{i \in S} \frac{1}{1 + d_S^-(i)}. \quad (5)$$

In particular, $F_G(\emptyset) = 0$. For a positive integer s , let $H_s = \sum_{j=1}^s 1/j$ be the s th harmonic number.

Lemma 3. *For every nonempty $S \subseteq [n]$, an independent set $I \subseteq S$ satisfying*

$$F_G(S) \leq H_{|S|}|I| \quad (6)$$

can be found in $O(n^2)$ time.

Proof. Process the vertices of S in increasing order. Assign vertex i the smallest positive integer that has not been assigned to any of its previously processed neighbors. Write $c(i)$ for its assigned color.

Vertex i has exactly $d_S^-(i)$ previously processed neighbors. These neighbors forbid at most $d_S^-(i)$ colors, so at least one of the integers $1, \dots, d_S^-(i) + 1$ is available. Therefore

$$c(i) \leq d_S^-(i) + 1, \quad \text{and hence} \quad \frac{1}{1 + d_S^-(i)} \leq \frac{1}{c(i)}.$$

If two vertices are adjacent, the later vertex avoids the color of the earlier one. The resulting coloring is thus proper.

Let k be the number of colors used and let

$$I_j = \{i \in S : c(i) = j\}, \quad j = 1, \dots, k.$$

First-fit uses the consecutive colors $1, \dots, k$, and every I_j is an independent set. Choose a largest class I , so that $|I_j| \leq |I|$ for every j . We obtain

$$F_G(S) = \sum_{i \in S} \frac{1}{1 + d_S^-(i)} \leq \sum_{i \in S} \frac{1}{c(i)} = \sum_{j=1}^k \sum_{i \in I_j} \frac{1}{j} = \sum_{j=1}^k \frac{|I_j|}{j} \leq |I| \sum_{j=1}^k \frac{1}{j} = |I| H_k \leq |I| H_{|S|}.$$

The last inequality follows from $k \leq |S|$. A direct implementation examines at most n earlier vertices and at most n colors for each vertex, giving the stated running time. $\square$

The order in this proof is essential: the coloring order must agree with the order in the definition of $d_S^-(i)$. The algorithm computes a greedy coloring and a largest class of that coloring; it does not compute a maximum independent set.

The harmonic factor in (6) cannot be improved for arbitrary ordered graphs. Indeed, for the complete graph K_s on $[s]$, every vertex i has $i - 1$ earlier neighbors. Thus

$$F_{K_s}([s]) = \sum_{i=1}^s \frac{1}{1 + (i-1)} = H_s, \quad \alpha(K_s) = 1.$$

This example shows sharpness of the graph inequality, rather than an exact approximation threshold for the assortment problem.

3 The reduction from Maximum Independent Set

We use the construction of Akchen and Mitrofanov [1, EC.3.3], which adapts the ranking-preference construction of Aouad et al. [2]. We give the details to make the approximation argument self-contained.

3.1 Construction and encoding length

Let $G = ([n], E)$ be a graph with $n \geq 2$. For each vertex i , introduce a product i and a customer type i . Set

$$B = n^2, \quad Z = \sum_{h=1}^n B^{-h}. \quad (7)$$

The consideration sets, revenues, and type probabilities are

$$C_i = \{i\} \cup \{j \in [n] : j < i \text{ and } \{i, j\} \in E\}, \quad r_i = ZB^i, \quad \lambda_i = \frac{B^{-i}}{Z}, \quad i \in [n]. \quad (8)$$

These parameters satisfy

$$\sum_{i=1}^n \lambda_i = \frac{1}{Z} \sum_{i=1}^n B^{-i} = 1, \quad \lambda_i r_i = \frac{B^{-i}}{Z} ZB^i = 1. \quad (9)$$

All probabilities and revenues are positive. Moreover, the maximum element of C_i is i , so the n consideration sets are pairwise distinct. We denote the resulting assortment instance by $\mathcal{I}(G)$, its revenue function by R_G , and its optimal revenue by OPT_G .

The revenues in (8) grow rapidly, but all parameters have binary representations of polynomial length. To see this explicitly, define the integer

$$Q = \sum_{h=0}^{n-1} B^h.$$

Multiplying Z by B^n and reversing the order of summation gives

$$B^n Z = \sum_{h=1}^n B^{n-h} = Q.$$

Consequently, the parameters can be written as ratios of integers:

$$\lambda_i = \frac{B^{n-i}}{Q}, \quad r_i = \frac{Q}{B^{n-i}}. \quad (10)$$

Since $B \geq 4$ and

$$Q = \frac{B^n - 1}{B - 1} < B^n,$$

the integers in (10) have $O(n \log n)$ bits each. There are $2n$ rational parameters and at most n^2 incidence entries for the consideration sets. The full instance therefore has encoding length $O(n^2 \log n)$ and can be constructed in polynomial time. The objective can also be evaluated exactly in polynomial time by rational arithmetic.

3.2 Relating revenue to the graph

The products in C_i other than i have smaller indices and much smaller revenues. After multiplication by the probability of type i , a purchase of product $j \in C_i$ contributes the factor

$$\lambda_i r_j = \frac{B^{-i}}{Z} Z B^j = B^{j-i}. \quad (11)$$

For $j = i$ this factor is one; for $j < i$ it is at most $1/B$. This separation allows us to identify the main part of the revenue.

Lemma 4. *For every assortment $S \subseteq [n]$ in $\mathcal{I}(G)$,*

$$R_G(S) = F_G(S) + \delta_G(S), \quad 0 \leq \delta_G(S) \leq \frac{1}{n}. \quad (12)$$

In addition, every independent set J in G satisfies $R_G(J) \geq |J|$. In particular, $\text{OPT}_G \geq \alpha(G)$.

Proof. Fix S and write $k_i = |C_i \cap S|$. We first consider a type i for which $k_i > 0$. Using (11), its contribution to expected revenue is

$$\lambda_i R_i(S) = \frac{1}{k_i} \sum_{j \in C_i \cap S} \lambda_i r_j = \frac{1}{k_i} \sum_{j \in C_i \cap S} B^{j-i} = \frac{\mathbf{1}\{i \in S\}}{k_i} + \frac{1}{k_i} \sum_{\substack{j \in C_i \cap S \\ j < i}} B^{j-i}.$$

The last equality separates the possible term $j = i$ from all terms with $j < i$. Here $\mathbf{1}\{i \in S\}$ is one when $i \in S$ and zero otherwise. The separation is valid because every member of C_i is at most i .

For each $i \in S$, the definition of C_i gives

$$k_i = 1 + |\{j \in S : j < i \text{ and } \{i, j\} \in E\}| = 1 + d_S^-(i).$$

Thus the first terms in the revenue calculation sum to

$$\sum_{\substack{i \in [n] \\ k_i > 0}} \frac{\mathbf{1}\{i \in S\}}{k_i} = \sum_{i \in S} \frac{1}{1 + d_S^-(i)} = F_G(S).$$

Define the remainder from type i by

$$\delta_i(S) = \begin{cases} \frac{1}{k_i} \sum_{\substack{j \in C_i \cap S \\ j < i}} B^{j-i}, & k_i > 0, \\ 0, & k_i = 0. \end{cases}$$

For every summand with $j < i$, we have $B^{j-i} \leq B^{-1}$. There are at most k_i such summands. Therefore, when $k_i > 0$,

$$0 \leq \delta_i(S) \leq \frac{1}{k_i} \sum_{\substack{j \in C_i \cap S \\ j < i}} \frac{1}{B} \leq \frac{1}{k_i} \frac{k_i}{B} = \frac{1}{B}.$$

The same bound holds when $k_i = 0$. Setting $\delta_G(S) = \sum_{i=1}^n \delta_i(S)$ now gives both the decomposition and the estimate

$$0 \leq \delta_G(S) \leq \sum_{i=1}^n \frac{1}{B} = \frac{n}{B} = \frac{1}{n}.$$

This sum includes every customer type. In particular, types with $i \notin S$ may still purchase a product and are included in $\delta_G(S)$.

For the second claim, let J be an independent set. If $i \in J$, none of the earlier neighbors of i belongs to J . Hence

$$C_i \cap J = \{i\}, \quad R_i(J) = r_i.$$

All other type contributions are nonnegative, so

$$R_G(J) = \sum_{i=1}^n \lambda_i R_i(J) \geq \sum_{i \in J} \lambda_i R_i(J) = \sum_{i \in J} \lambda_i r_i = \sum_{i \in J} 1 = |J|.$$

Choosing J to be a maximum independent set yields $\text{OPT}_G \geq R_G(J) \geq \alpha(G)$. $\square$

Combining the two lemmas gives the required independent-set extraction. For every nonempty S , Lemma 3 returns a nonempty independent set $I \subseteq S$ for which

$$R_G(S) = F_G(S) + \delta_G(S) \leq H_{|S|}|I| + \frac{1}{n} \leq H_n|I| + \frac{1}{n} \leq \left(H_n + \frac{1}{n}\right)|I|. \quad (13)$$

The last step uses $|I| \geq 1$. It turns the additive remainder into a multiplicative factor without subtracting it from the approximation guarantee. Also, since $|I| \leq \alpha(G)$, the preceding estimates imply

$$\alpha(G) \leq \text{OPT}_G \leq H_n \alpha(G) + \frac{1}{n}. \quad (14)$$

The upper bound applies to the empty assortment as well, whose revenue is zero.

4 Transferring the approximation guarantee

We first state the reduction's guarantee for a general approximation factor. For Maximum Independent Set, a factor $\gamma(n) \geq 1$ means that the algorithm returns an independent set of size at least $\alpha(G)/\gamma(n)$ on every n -vertex graph.

Proposition 5. *A deterministic polynomial-time $\rho(n)$ -approximation algorithm for assortment optimization under the consideration set model gives a deterministic polynomial-time $\rho(n)(H_n + 1/n)$ -approximation algorithm for Maximum Independent Set.*

Proof. Given a graph G with $n \geq 2$, construct $\mathcal{I}(G)$ and run the assortment algorithm to obtain S . By its approximation guarantee and Lemma 4,

$$R_G(S) \geq \frac{\text{OPT}_G}{\rho(n)} \geq \frac{\alpha(G)}{\rho(n)}. \quad (15)$$

Since $n \geq 2$, we have $\alpha(G) \geq 1$, so the right-hand side is positive. As $R_G(\emptyset) = 0$, the returned assortment is nonempty.

Apply first-fit coloring to $G[S]$ in the order inherited from G , and return a largest color class I . Combining (13) and (15) gives

$$\left(H_n + \frac{1}{n}\right) |I| \geq R_G(S) \geq \frac{\alpha(G)}{\rho(n)}.$$

Dividing by the positive quantity $H_n + 1/n$ yields

$$|I| \geq \frac{\alpha(G)}{\rho(n)(H_n + 1/n)}.$$

The construction has polynomial encoding length, and the extraction takes $O(n^2)$ time. The resulting graph algorithm is therefore polynomial time. A graph with at most one vertex is solved directly. $\square$

We use Zuckerman's deterministic inapproximability result for Maximum Clique [3, Theorem 1.1]: for every fixed $\eta \in (0, 1)$, a polynomial-time $n^{1-\eta}$ -approximation for that problem implies $P = NP$. An independent set in G is exactly a clique in its complement $\overline{G}$. Complementation is polynomial time and preserves the vertex count. The same hardness statement therefore holds for Maximum Independent Set.

Proof of Theorem 1. Suppose that, for some fixed $\varepsilon \in (0, 1)$, the assortment problem has a deterministic polynomial-time $O(n^{1-\varepsilon})$ -approximation algorithm. There exist constants $a \geq 1$ and n_1 such that its approximation factor is at most $an^{1-\varepsilon}$ whenever $n \geq n_1$. By Proposition 5, this algorithm yields an independent set of size at least

$$\frac{\alpha(G)}{an^{1-\varepsilon}(H_n + 1/n)} \quad (16)$$

on every n -vertex graph with $n \geq \max\{2, n_1\}$.

The harmonic numbers satisfy $H_n \leq 1 + \log n$, where $\log$ denotes the natural logarithm. Since ε and a are fixed,

$$\lim_{n \rightarrow \infty} \frac{a(1 + \log n + 1/n)}{n^{\varepsilon/2}} = 0.$$

There is consequently an integer $n_0 \geq \max\{2, n_1\}$, depending only on these constants, such that

$$a \left(H_n + \frac{1}{n}\right) \leq n^{\varepsilon/2} \quad \text{for every } n \geq n_0.$$

For these values of n , the denominator in (16) is at most

$$an^{1-\varepsilon} \left(H_n + \frac{1}{n}\right) \leq n^{1-\varepsilon} n^{\varepsilon/2} = n^{1-\varepsilon/2}.$$

Thus the extracted independent set satisfies

$$|I| \geq \frac{\alpha(G)}{n^{1-\varepsilon/2}}.$$

For $n < n_0$, compute a maximum independent set by exhaustive search. Because n_0 is a constant independent of the input graph, this produces a polynomial-time $n^{1-\varepsilon/2}$ -approximation for Maximum Independent Set on all graphs. Zuckerman’s theorem with $\eta = \varepsilon/2$ then implies $P = NP$.

Finally, the construction uses exactly n pairwise distinct consideration sets and positive rational parameters. Hence the argument applies under the restrictions stated in Theorem 1. $\square$

5 Discussion

The improvement comes entirely from extracting an independent set from the existing assortment instance. The earlier argument selects customer types i whose conditional expected revenue satisfies $R_i(S) \geq r_i/\sqrt{n}$, where r_i is the revenue of their own product [1, EC.3.3]. The first-fit argument instead groups all offered vertices by color and sums their contributions with weights $1, 1/2, \dots, 1/k$. This gives a harmonic loss while keeping the construction and the number of products unchanged.

Theorem 1 and Proposition 2 determine the polynomial exponent of the worst-case approximation factor. They do not exclude an approximation factor such as $n/\log n$. The complete-graph example in Section 2 shows that the harmonic bound on F_G is sharp, but does not settle whether a different algorithm or reduction can improve the remaining bounds for assortment optimization.

The numerical assumptions also matter. In the constructed instance,

$$\frac{r_n}{r_1} = B^{n-1} = n^{2(n-1)}.$$

The reduction therefore uses a large revenue ratio and correspondingly small customer-type probabilities. Polynomial binary encoding suffices for the stated hardness result, but this argument does not establish strong NP-hardness or the same approximation bound under a restriction on the revenue ratio. Throughout, the excluded algorithms are deterministic; excluding randomized approximation guarantees would require an appropriate randomized complexity assumption.

Use of AI tools. OpenAI’s ChatGPT and Codex (GPT-6) were used to develop the proof and prepare the manuscript. The human author takes full responsibility for the mathematical arguments and the final manuscript.

References

- [1] Yi-Chun Akchen and Dmitry Mitrofanov. Consider or choose? The role and power of consideration sets. *Management Science*, 72(7):6025–6042, 2026. doi:10.1287/mnsc.2023.00442. See also the electronic companion, Section EC.3.3, and arXiv:2302.04354v4.
- [2] Ali Aouad, Vivek Farias, Retsef Levi, and Danny Segev. The approximability of assortment optimization under ranking preferences. *Operations Research*, 66(6):1661–1669, 2018. doi:10.1287/opre.2018.1754.
- [3] David Zuckerman. Linear degree extractors and the inapproximability of Max Clique and Chromatic Number. *Theory of Computing*, 3(6):103–128, 2007. doi:10.4086/toc.2007.v003a006.